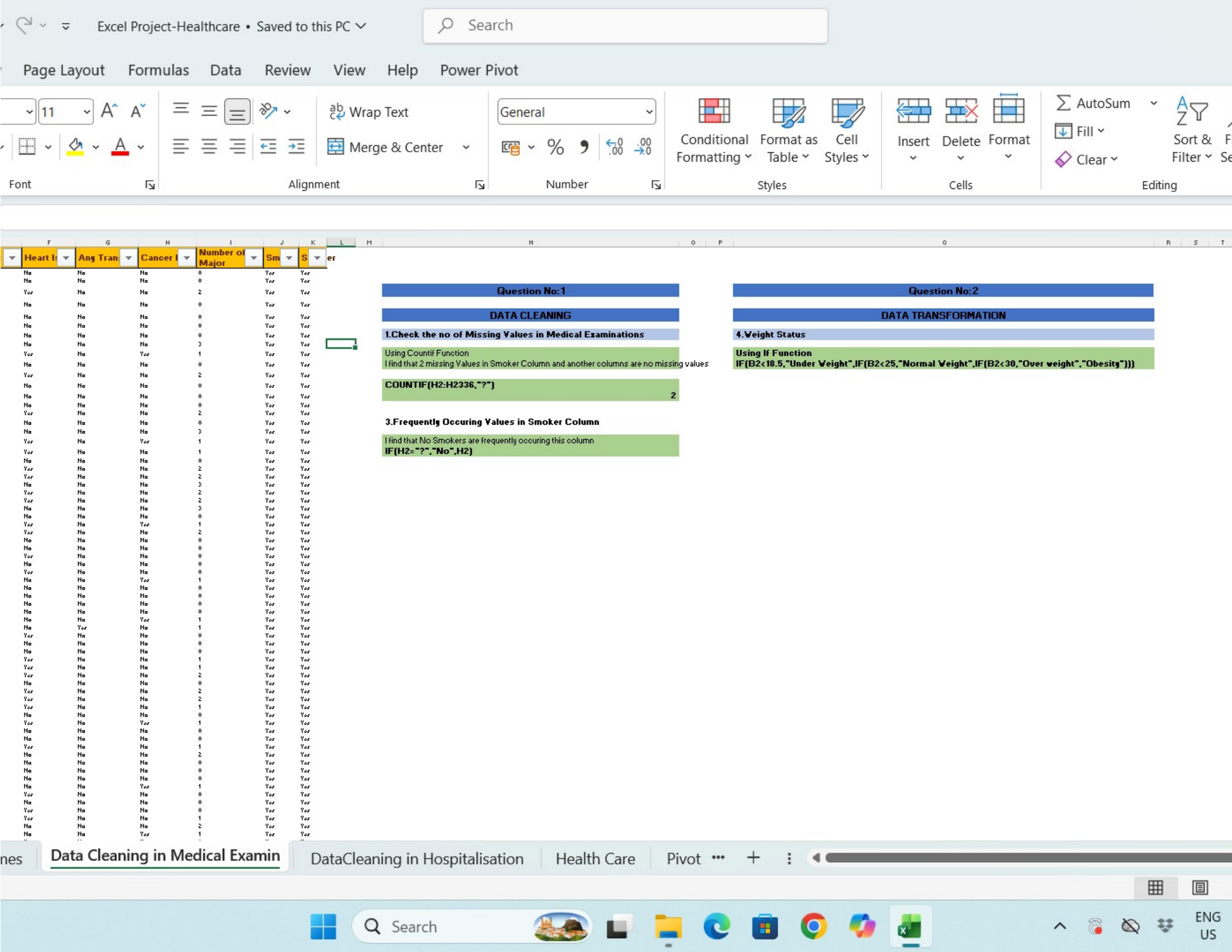

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Question No:1
DATA CLEANING
1.Check the no of Missing Values in Medical Examinations
Using Countif Function
I find that 2 missing Values in Smoker Column and another columns are no missing values
COUNTIF(H2:H2336,"?")
2
3.Frequently Occuring Values in Smoker Column
I find that No Smokers are frequently occuring this column
IF(H2="?","No",H2)
Question No:2
DATA TRANSFORMATION
4.Weight Status
Using If Function
IF(B2<18.5,"Under Weight",IF(B2<25,"Normal Weight",IF(B2<30,"Over weight","Obesity")))
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Question No:2

DATA TRANSFORMATION

1.Split the Columns in Customer Names

Using Text to Column Delimiter to separate the three columns

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Year

DD-MM-YY	Age	Children	Charges	Hospital tier	City tier	State ID
9-Jul-1992	31	0	$563.84	tier - 2	tier - 3	R1013
30-Nov-1992	31	0	$570.62	tier - 2	tier - 1	R1013
30-Jun-1993	30	0	$600.00	tier - 2	tier - 1	R1013
13-Sep-1992	31	0	$604.54	tier - 3	tier - 3	R1013
27-Jul-1998	25	0	$637.26	tier - 3	tier - 3	R1013
20-Nov-2001	22	0	$646.14	tier - 3	tier - 3	R1012
1-Jun-1993	30	0	$650.00	tier - 3	tier - 3	R1013
4-Jul-1995	28	0	$650.00	tier - 3	tier - 3	R1013
29-Nov-2002	21	0	$668.00	tier - 3	tier - 2	R1012
9-Nov-1997	26	0	$670.00	tier - 3	tier - 3	R1013
12-Sep-2001	22	0	$687.54	tier - 3	tier - 2	R1013
26-Dec-1999	24	0	$700.00	tier - 2	tier - 3	R1013
14-Dec-1999	24	0	$722.99	tier - 3	tier - 1	R1013
19-Sep-2002	21	0	$750.00	tier - 3	tier - 1	R1012
9-Aug-1993	30	0	$760.00	tier - 3	tier - 1	R1013
22-Oct-1996	27	0	$760.00	tier - 3	tier - 3	R1013
28-Jun-1993	30	0	$770.00	tier - 3	tier - 3	R1013
18-Sep-1996	27	0	$770.38	tier - 3	tier - 2	R1012
7-Dec-1995	28	0	$773.54	tier - 3	tier - 2	R1013
7-Oct-2004	19	0	$830.52	tier - 3	tier - 2	R1011
18-Nov-2000	23	0	$865.41	tier - 3	tier - 1	R1013
27-Nov-1993	30	0	$896.21	tier - 3	tier - 1	R1013
30-Oct-1994	29	0	$915.07	tier - 3	tier - 1	R1013
28-Oct-1995	28	0	$928.59	tier - 3	tier - 1	R1013
19-Aug-2001	22	0	$964.71	tier - 3	tier - 2	R1013
22-Nov-1994	29	0	$1,006.65	tier - 3	tier - 2	R1013
9-Oct-2000	23	0	$1,012.00	tier - 3	tier - 2	R1013
5-Aug-1999	24	0	$1,019.00	tier - 3	tier - 2	R1013
5-Oct-1995	28	0	$1,033.74	tier - 3	tier - 1	R1013
17-Nov-1994	29	0	$1,044.00	tier - 3	tier - 2	R1013
20-Jul-2002	21	0	$1,047.00	tier - 3	tier - 1	R1013
1-Jul-2002	21	0	$1,049.00	tier - 3	tier - 2	R1012
1-Jun-1995	28	0	$1,056.00	tier - 3	tier - 2	R1013
28-Sep-1993	30	0	$1,068.00	tier - 3	tier - 1	R1013
24-Aug-2004	19	0	$1,070.00	tier - 3	tier - 3	R1013
13-Aug-1997	26	0	$1,071.00	tier - 3	tier - 3	R1013
26-Dec-1998	25	0	$1,082.00	tier - 3	tier - 2	R1013
13-Aug-2002	21	0	$1,086.00	tier - 3	tier - 3	R1013

Question No:1

DATA CLEANING

1.Check the no of Missing Values in Hospitalisation Details

Using Countif Function
I find that Missing Values are there in six columns

Customer Id

COUNTIF(A2:A2344,"?")

6

Using Unknown in ? Places.

Year

COUNTIF(B2:B2344,"?")

2

Month

COUNTIF(C2:C2344,"?")

3

Date,Children and Charges columns are there is no missing values

Hospital tier

COUNTIF(G2:G2344,"?")

1

City tier

COUNTIF(H2:H2344,"?")

1

State ID

COUNTIF(I2:I2344,"?")

1.Convert the Numerical D

Using Find and Replace fun

2.Check the inconsistent

No Inconsistent Data in th

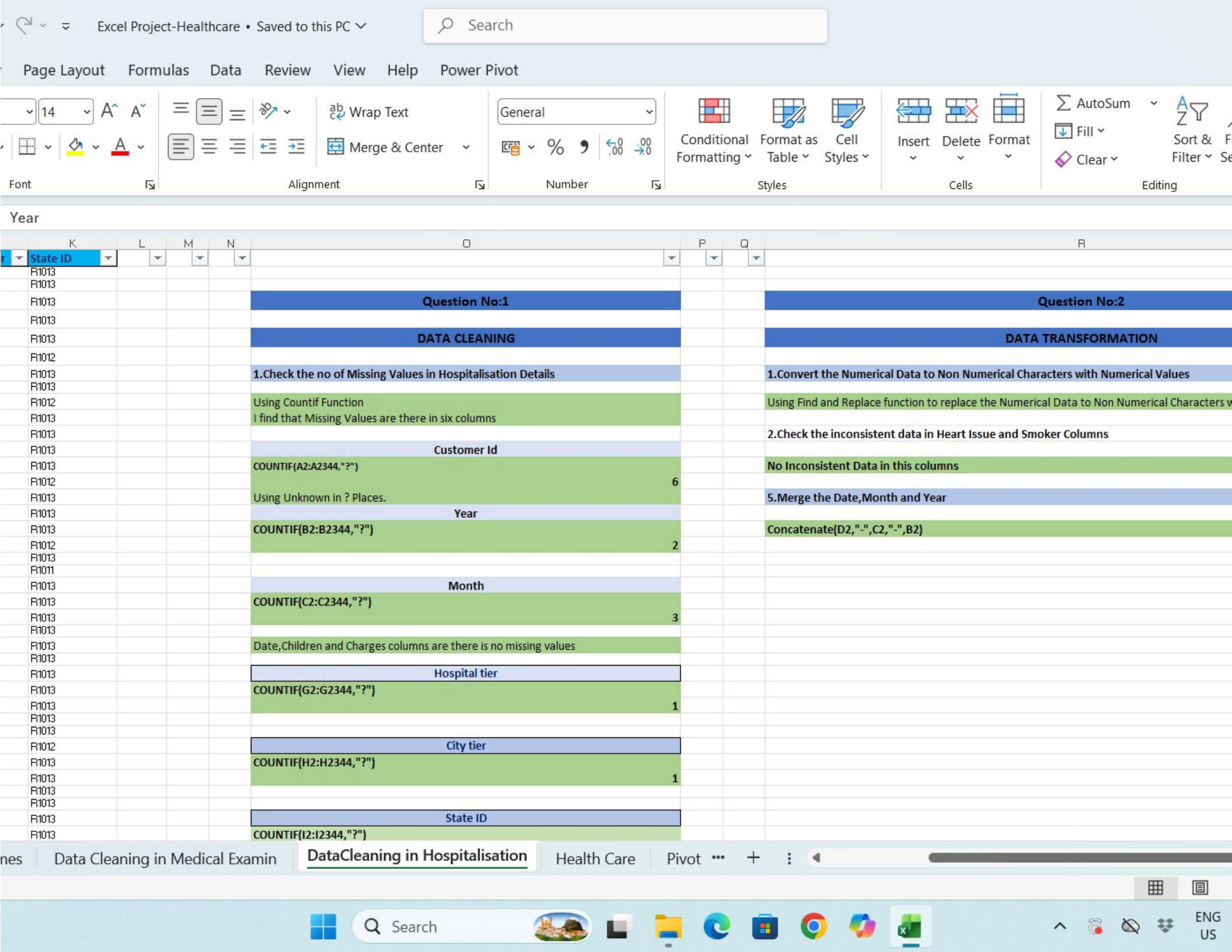
5.Merge the Date,Month

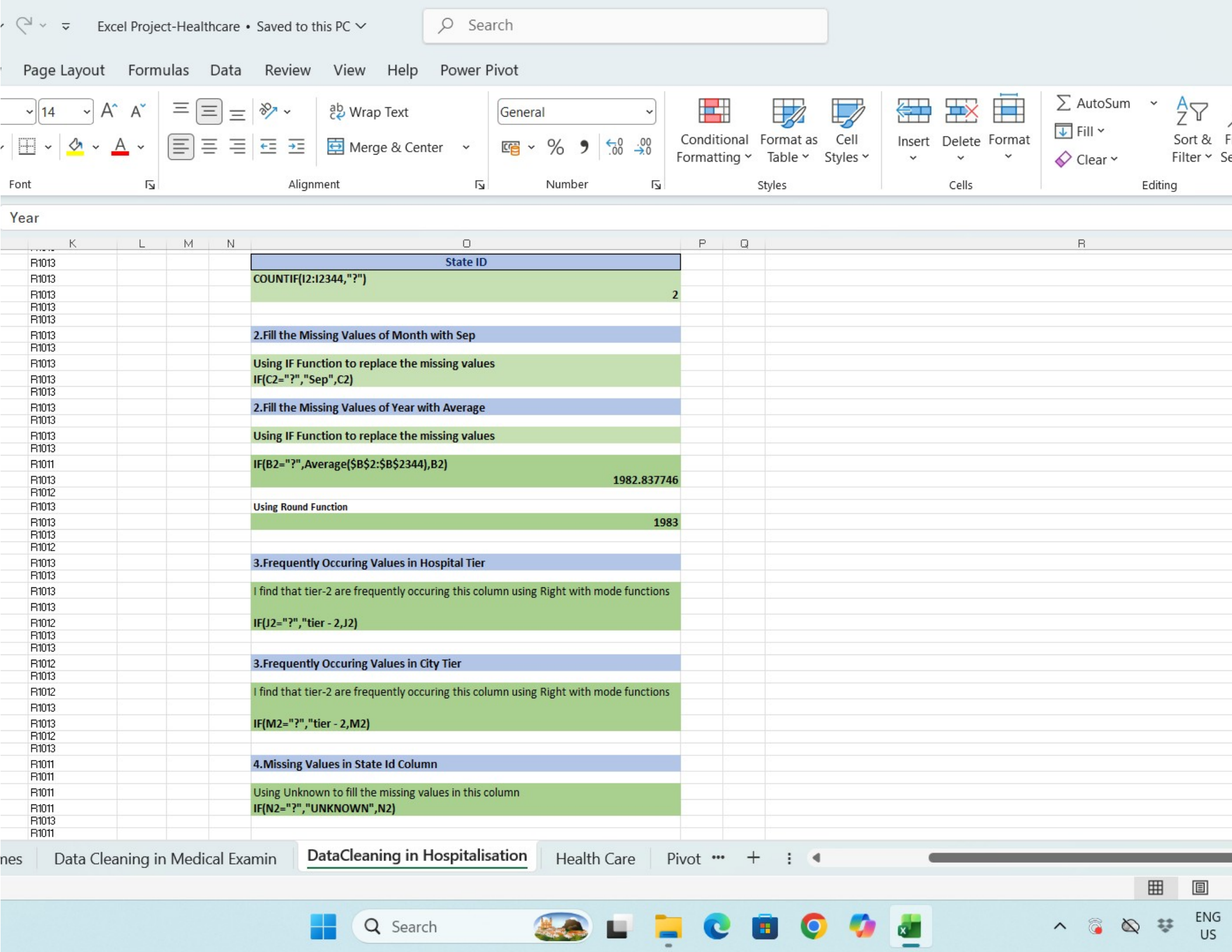
Concatenate(D2,"-",C2,"-

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Question No:1
DATA CLEANING
1.Check the no of Missing Values in Hospitalisation Details
Using Countif Function
I find that Missing Values are there in six columns
Customer Id
COUNTIF(A2:A2344,"?")
6
Using Unknown in ? Places.
Year
COUNTIF(B2:B2344,"?")
2
Month
COUNTIF(C2:C2344,"?")
3
Date,Children and Charges columns are there is no missing values
Hospital tier
COUNTIF(G2:G2344,"?")
1
City tier
COUNTIF(H2:H2344,"?")
1
State ID
COUNTIF(I2:I2344,"?")
Question No:2
DATA TRANSFORMATION
1.Convert the Numerical Data to Non Numerical Characters with Numerical Values
Using Find and Replace function to replace the Numerical Data to Non Numerical Characters w
2.Check the inconsistent data in Heart Issue and Smoker Columns
No Inconsistent Data in this columns
5.Merge the Date,Month and Year
Concatenate(D2,"-",C2,"-",B2)
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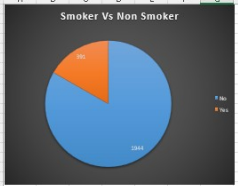

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Cells
Editing
Year
State ID
COUNTIF(I2:I2344,"?")
2
2.Fill the Missing Values of Month with Sep
Using IF Function to replace the missing values
IF(C2="?","Sep",C2)
2.Fill the Missing Values of Year with Average
Using IF Function to replace the missing values
IF(B2="?",Average($B$2:$B$2344),B2)
1982.837746
Using Round Function
1983
3.Frequently Occuring Values in Hospital Tier
I find that tier-2 are frequently occuring this column using Right with mode functions
IF(J2="?","tier - 2,J2)
3.Frequently Occuring Values in City Tier
I find that tier-2 are frequently occuring this column using Right with mode functions
IF(M2="?","tier - 2,M2)
4.Missing Values in State Id Column
Using Unknown to fill the missing values in this column
IF(N2="?","UNKNOWN",N2)
Data Cleaning in Medical Examin
DataCleaning in Hospitalisation
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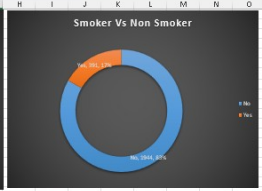
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Any Transplants	Cancer History	Number of Major Surgeries	Smoker	Weight Status	Diabetes Status	Date of Birth	Charges	Hospital Tier	City Tier	State ID	Age
No	No	0	Yes	Obesity	Diabetes	12-Oct-1968	$63,770.43	tier - 1	tier - 3	R1013	55
No	No	0	Yes	Obesity	Prediabetes	8-Jun-1977	$62,592.87	tier - 2	tier - 3	R1013	46
No	No	2	Yes	Obesity	Diabetes	11-Sep-1970	$60,021.40	tier - 1	tier - 1	R1012	53
No	No	0	Yes	Obesity	Prediabetes	6-Jun-1991	$58,571.07	tier - 1	tier - 3	R1024	32
No	No	0	Yes	Obesity	Normal	19-Jun-1989	$55,135.40	tier - 1	tier - 2	R1012	34
No	No	0	Yes	Obesity	Diabetes	4-Aug-1962	$52,590.83	tier - 1	tier - 3	R1011	61
No	No	0	Yes	Obesity	Prediabetes	27-Oct-1994	$51,194.56	tier - 1	tier - 3	R1011	29
No	No	3	Yes	Obesity	Diabetes	27-Jun-1958	$49,577.66	tier - 2	tier - 2	R1013	65
No	Yes	1	Yes	Obesity	Diabetes	4-Sep-1963	$48,970.25	tier - 1	tier - 2	R1013	60
No	No	0	Yes	Obesity	Diabetes	29-Dec-1978	$48,885.14	tier - 1	tier - 2	R1013	45
No	No	2	Yes	Obesity	Prediabetes	22-Jul-1959	$48,824.45	tier - 2	tier - 1	R1011	64
No	No	0	Yes	Obesity	Diabetes	27-Oct-1965	$48,675.52	tier - 1	tier - 2	R1013	58
No	No	0	Yes	Obesity	Diabetes	11-Oct-1962	$48,673.56	tier - 1	tier - 2	R1013	61
No	No	0	Yes	Obesity	Diabetes	1-Dec-1968	$48,549.18	tier - 1	tier - 3	R1016	55
No	No	2	Yes	Obesity	Diabetes	21-Dec-1961	$48,517.56	tier - 1	tier - 3	R1024	62
No	No	0	Yes	Obesity	Diabetes	27-Aug-1962	$48,173.36	tier - 1	tier - 3	R1011	61
No	No	3	Yes	Obesity	Diabetes	16-Nov-1958	$47,928.03	tier - 2	tier - 3	R1011	65
No	Yes	1	Yes	Obesity	Diabetes	5-Aug-1963	$47,896.79	tier - 1	tier - 3	R1024	60
No	No	1	Yes	Obesity	Normal	7-Nov-1964	$47,496.49	tier - 1	tier - 3	R1012	59
No	No	0	Yes	Obesity	Diabetes	27-Sep-1971	$47,462.89	tier - 1	tier - 2	R1013	52
No	No	2	Yes	Obesity	Diabetes	13-Jun-1961	$47,403.88	tier - 1	tier - 3	R1011	62
No	No	2	Yes	Obesity	Prediabetes	4-Oct-1959	$47,305.31	tier - 2	tier - 1	R1011	64
No	No	3	Yes	Obesity	Diabetes	3-Sep-1958	$47,291.06	tier - 2	tier - 1	R1011	65
No	No	2	Yes	Obesity	Diabetes	28-Dec-1970	$47,269.85	tier - 1	tier - 2	R1013	53
No	No	2	Yes	Obesity	Normal	14-Jul-1959	$47,055.53	tier - 2	tier - 1	R1013	64
No	No	3	Yes	Obesity	Diabetes	15-Sep-1958	$46,889.26	tier - 2	tier - 3	R1013	65
No	No	0	Yes	Obesity	Diabetes	17-Nov-1960	$46,718.16	tier - 1	tier - 1	R1012	63
No	Yes	1	Yes	Obesity	Normal	23-Jun-1969	$46,661.44	tier - 1	tier - 3	R1012	54
No	No	2	Yes	Obesity	Diabetes	8-Aug-1961	$46,599.11	tier - 1	tier - 1	R1013	62
No	No	0	Yes	Obesity	Diabetes	6-Dec-1971	$46,255.11	tier - 1	tier - 3	R1024	52
No	No	0	Yes	Obesity	Diabetes	17-Aug-1978	$46,200.99	tier - 1	tier - 1	R1013	45
No	No	0	Yes	Obesity	Normal	2-Dec-1976	$46,151.12	tier - 1	tier - 3	R1013	47
No	No	0	Yes	Obesity	Diabetes	4-Dec-1962	$46,130.53	tier - 1	tier - 1	R1012	61
No	No	0	Yes	Obesity	Prediabetes	30-Sep-1985	$46,113.51	tier - 2	tier - 3	R1011	38
No	Yes	1	Yes	Obesity	Prediabetes	12-Aug-1979	$45,863.21	tier - 2	tier - 1	R1013	44
No	No	0	Yes	Obesity	Diabetes	21-Dec-1960	$45,710.21	tier - 2	tier - 2	R1016	63
No	No	0	Yes	Obesity	Diabetes	9-Aug-1974	$45,702.02	tier - 2	tier - 2	R1012	49
No	No	0	Yes	Obesity	Diabetes	17-Dec-1962	$45,008.96	tier - 2	tier - 2	R1013	61
No	No	0	Yes	Obesity	Diabetes	13-Aug-1971	$44,641.20	tier - 2	tier - 3	R1024	52
No	Yes	1	Yes	Obesity	Normal	11-Oct-1993	$44,585.46	tier - 2	tier - 2	R1011	30
Yes	No	1	Yes	Obesity	Normal	2-Aug-2000	$44,501.40	tier - 2	tier - 3	R1013	23
No	No	0	Yes	Obesity	Diabetes	14-Sep-1967	$44,423.80	tier - 2	tier - 3	R1013	56
No	No	0	Yes	Obesity	Diabetes	29-Oct-1971	$44,400.41	tier - 2	tier - 2	R1013	52
No	No	0	Yes	Obesity	Diabetes	24-Nov-1968	$44,260.75	tier - 2	tier - 1	R1013	55
No	No	1	Yes	Obesity	Diabetes	1-Jun-1975	$44,202.65	tier - 2	tier - 2	R1013	48

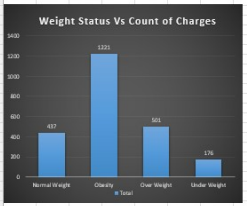
Row Labels	Count of Charges
Id1	1
tier - 1	1
No Major Surg	1
tier - 3	1
No	1
No	1
Id10	1
tier - 1	1
No Major Surg	1
tier - 3	1
No	1
No	1
Id100	1
tier - 1	1
No Major Surg	1
tier - 2	1
No	1
No	1
Id1000	1
tier - 3	1
No Major Surg	1
tier - 2	1
No	1
No	1
Id1001	1
tier - 3	1
1 Major Surger	1
tier - 2	1
No	1
Yes	1
Id1002	1
tier - 3	1
No Major Surg	1
tier - 2	1
No	1
Yes	1
Id1003	1
tier - 3	1
2 Major Surger	1
tier - 1	1
No	1
Yes	1
Id1004	1
tier - 3	1

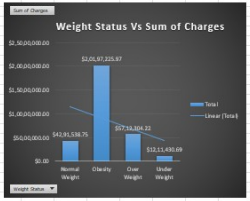
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Row Lab	Count of Smoker
No	1344
Yes	331
Grand Tot	2335



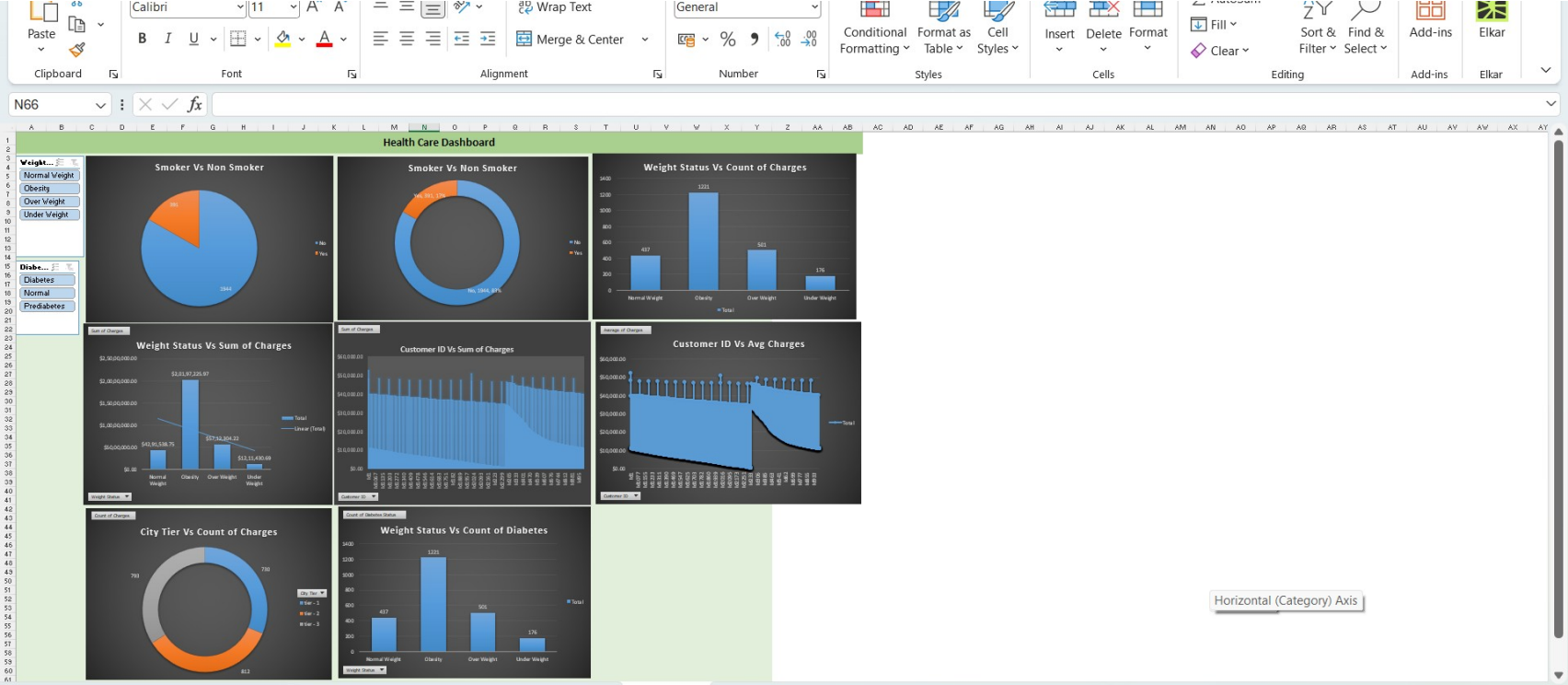


Row Lab	Count of Charges
Under Weight	176
Grand Tot	176

Row Lab	Sum of Charges
Normal Weight	$42,91,536.75
Obesity	$2,01,97,225.97
Over Weight	$57,12,304.22
Under Weight	$12,11,430.69
Grand Total	**********

- I am Create the Pivot Table with Charts also..
- I analyse from the raw data to clean and without error in this data set
- I am using the Pie and Donut Chart for cancer distribution among Smoker Vs Non Smoker
- I am using the Line Chart for Weight Status Vs Sum And count of Charges

Chart Area



Horizontal (Category) Axis